UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

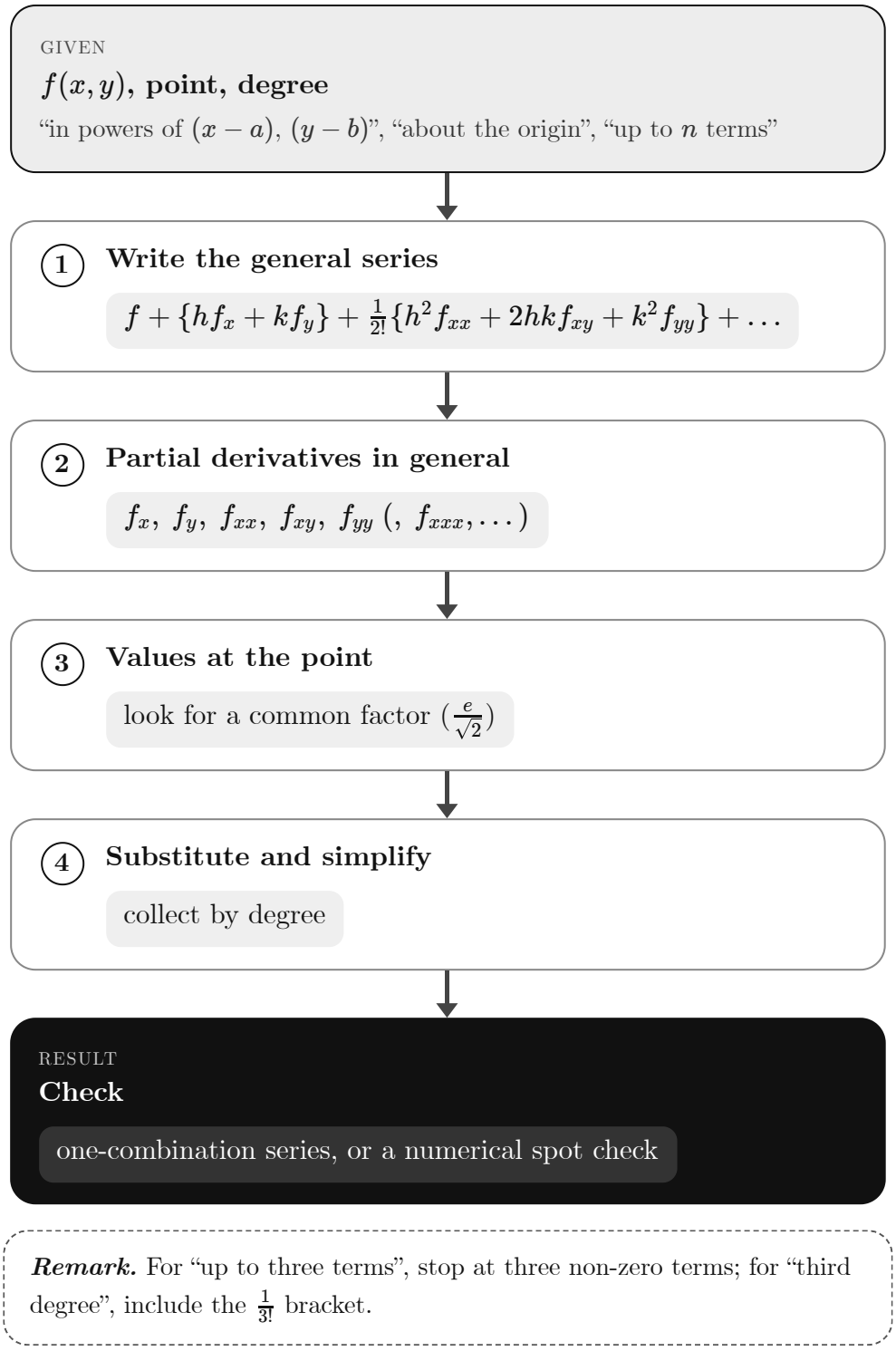
Class 008 · Problems on Taylor's and Maclaurin's series · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S008·12	Deck 008, slide 12 (rendered .pptx)
T008 00:07:38	AV-summary transcript 008, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b

Contents

1. Figure F1.1 · Solving a Taylor or Maclaurin expansion problem
2. Figure F3.1 · Approximate value from a Taylor polynomial
3. Quick checks with a one-variable series

1 Taylor expansion about a point (a, b)



Remark. For "up to three terms", stop at three non-zero terms; for "third degree", include the $\frac{1}{3!}$ bracket.

Figure F1.1. Solving a Taylor or Maclaurin expansion problem ESA ★★★

T008 00:00:21 · T008 00:02:07 · T008 00:05:19

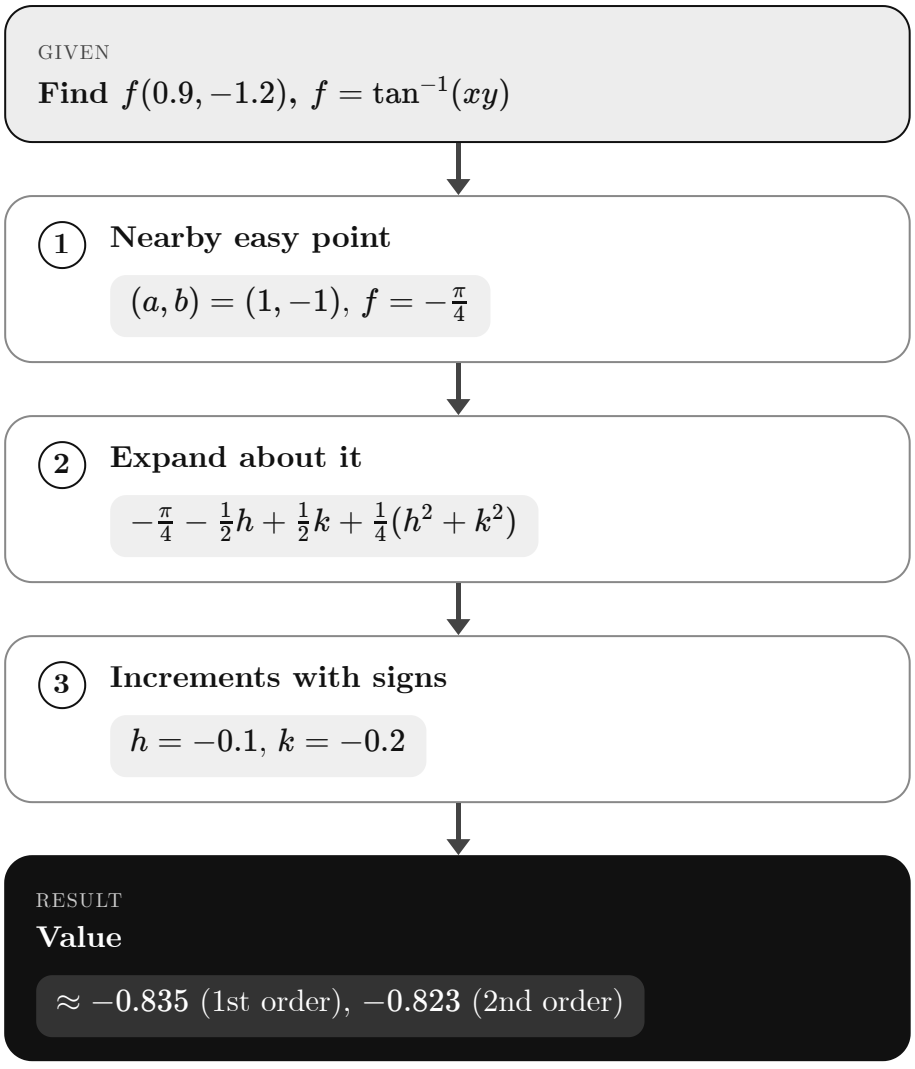
2 Maclaurin expansion about the origin

Quick checks with a one-variable series

Function	Combination t	One-variable series	Problem
$\log(2x+y+1)$	$2x+y$	$t-\frac{t^2}{2}+\frac{t^3}{3}-\dots$	W2.1
e^{x-y}	$x-y$	$1+t+\frac{t^2}{2}+\frac{t^3}{6}+\dots$	W2.2
$\log(1+x-y)$	$x-y$	$t-\frac{t^2}{2}+\frac{t^3}{3}-\dots$	W2.4
$\sin(x+2y)$	$x+2y$	$t-\frac{t^3}{6}+\dots$	W2.6
$e^x\log(1+y)$, $e^{ax}\sin by$	product	multiply the two series, keep degree ≤ 3	W2.3, W2.5

ADDED FOR CLARITY S008·6 · S008·11 · ESA Jan-24 s1 1b · ESA Jun-24 1c

3 Approximate values from a Taylor polynomial



Remark. Exact value -0.8238: the second-order term is worth keeping.

Figure F3.1. Approximate value from a Taylor polynomial ESA ★★

S008·7 · S008·9 · T008 00:12:10